

University of California, Davis
Department of Economics

PRELIMINARY EXAMINATION FOR THE Ph.D. DEGREE,
MACROECONOMICS

Answer Key: 200E Questions

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Question 3 (20 points)

Consider the planner's problem for a real business cycle model with inelastic labor supply and no trend growth. Time is infinite, and the planner seeks to maximize lifetime expected utility from consumption (C):

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} \right) \quad (1)$$

where $0 < \beta < 1$ and $\sigma > 0$.

In this model there are two vintages of capital, “new” K_1 and “old” K_2 , that are used to produce output, Y , according to a constant elasticity of substitution production technology:

$$Y_t = A_t \left(\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2) N^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (2)$$

where $\alpha_1 > 0$, $\alpha_2 > 0$ and $\alpha_1 + \alpha_2 \leq 1$. Labor, N , is inelastically supplied, so we can assume $N = 1$.¹

Capital lasts for two periods. New capital available at the start of period $t + 1$ (denoted $K_{1,t+1}$) is created by investment, I_t :

$$K_{1,t+1} = I_t$$

New capital depreciates at rate δ and becomes “old”, where $0 < \delta \leq 1$. In any period t , denote the old vintage of capital at the start of time t as $K_{2,t}$. The relationship between the two vintages is:

$$K_{2,t+1} = (1 - \delta) K_{1,t} \quad (3)$$

A higher value of δ implies a smaller fraction of the new capital stock today will be available tomorrow once it becomes “old” capital.

Total Factor Productivity, A_t , is stochastic. Assume this follows a stationary Markov process with no persistence. In other words, A_t is independent and identically distributed (i.i.d.).

The aggregate resource constraint is $Y_t = C_t + I_t$.

¹Note, the standard model studied in class with full depreciation ($\delta = 1$) and log utility has $\sigma \rightarrow 1$ (for simplicity, the utility function above omits the -1 from the numerator) and $\alpha_2 = 0$.

a) Write down the planner's problem in recursive form.

Answer:

The state variables are $K_{1,t}$, $K_{2,t}$, and A_t . The recursive formulation is:

$$V(K_{1,t}, K_{2,t}, A_t) = \max_{K_{1,t+1}, C_t} \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t [V(K_{1,t+1}, K_{2,t+1}, A_{t+1})] \right\}$$

Subject to:

$$\begin{aligned} Y_t &= A_t [\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2)]^{\frac{1}{1-\sigma}} \\ C_t + I_t &= Y_t \\ K_{1,t+1} &= I_t \\ K_{2,t+1} &= (1 - \delta) K_{1,t} \end{aligned}$$

or more compactly:

$$V(K_{1,t}, K_{2,t}, A_t) = \max_{K_{1,t+1}} \frac{\left(A_t [\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 ((1 - \delta) K_{1,t})^{1-\sigma} + (1 - \alpha_1 - \alpha_2)]^{\frac{1}{1-\sigma}} - K_{1,t+1} \right)^{1-\sigma}}{1 - \sigma} + \beta \mathbb{E}_t [V(K_{1,t+1}, (1 - \delta) K_{1,t}, A_{t+1})]$$

b) Derive the first order conditions for the planner's problem. Provide some intuition for the Euler equation you derive. (**Hint:** you will need two envelope conditions for the two types of capital).

Answer:

Let λ be the Lagrange multiplier on the resource constraint. The Lagrangian is:

$$\begin{aligned} \mathcal{L} &= \frac{C_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}[V(K_{1,t+1}, (1 - \delta) K_{1,t}, A_{t+1})] \\ &\quad + \lambda_t \left(A_t (\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2))^{\frac{1}{1-\sigma}} - C_t - K_{1,t+1} \right) \end{aligned}$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial C_t} = C_t^{-\sigma} - \lambda_t = 0 \Rightarrow \lambda_t = C_t^{-\sigma}$$

$$\frac{\partial \mathcal{L}}{\partial K_{1,t+1}} = -\lambda_t + \beta \mathbb{E} \left[\frac{\partial V(K_{1,t+1}, (1-\delta)K_{1,t}, A_{t+1})}{\partial K_{1,t+1}} \right] = 0$$

Envelope conditions:

$$\begin{aligned} \frac{\partial V}{\partial K_{1,t}} &= \lambda_t A_t (\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2))^{-\frac{\sigma}{1-\sigma}} \cdot \alpha_1 K_{1,t}^{-\sigma} \\ &\quad + \beta(1-\delta) \mathbb{E} \left[\frac{\partial V(K_{1,t+1}, (1-\delta)K_{1,t}, A_{t+1})}{\partial K_{2,t+1}} \right] \end{aligned}$$

$$\frac{\partial V}{\partial K_{2,t}} = \lambda_t A_t (\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2))^{-\frac{\sigma}{1-\sigma}} \alpha_2 K_{2,t}^{-\sigma}$$

Which yields an Euler equation:

$$C_t^{-\sigma} = \beta E \left[\alpha_1 A_{t+1}^{1-\sigma} \left(\frac{Y_{t+1}}{K_{1,t+1}} \right)^{\sigma} C_{t+1}^{-\sigma} + \beta(1-\delta) C_{t+2}^{-\sigma} \alpha_2 A_{t+2}^{1-\sigma} \left(\frac{Y_{t+2}}{K_{2,t+2}} \right)^{\sigma} \right]$$

Interpretation: The Euler equation balances the marginal utility cost of investing today with the expected marginal benefit of increased output from both new and old capital in the future. Note that the first part of the right hand side is very conventional and reflects the marginal product of capital for K_1 . Without the second term, this would say that expected consumption growth is related to the marginal product of capital in $t+1$. The first term reflects how one unit of K_1 directly raises output. The second term is the marginal product of old capital. Since new capital becomes old capital, the second term is the indirect effect of an additional unit of K_1 once it becomes old.

c) Solve this model using guess and verify. In particular, find the solved policy functions for output, consumption and investment.

(Hints: Guess that investment is a constant share of output and follow the usual approach. In addition, note that the marginal product for each type of capital $j \in (1, 2)$ takes the form: $\frac{\partial Y}{\partial K_j} = \alpha_j A^{1-\sigma} \left(\frac{Y}{K_j} \right)^{\sigma}$. Also assume $E(A_{t+1}^{1-\sigma} | A_t) = \mu$ where $\mu > 0$ is a fixed constant.)

Answer:

Guess: $I_t = \theta Y_t$. We therefore have the following set of policy functions:

$$C_t = (1 - \theta)Y_t, \quad K_{1,t+1} = \theta Y_t, \quad K_{2,t+1} = (1 - \delta)K_{1,t}$$

Plug these into the Euler from above:

$$\frac{(1-\theta)^{-\sigma}}{Y_t^\sigma} = \beta E \left[\alpha_1 A_{t+1}^{1-\sigma} \left(\frac{Y_{t+1}}{\theta Y_t} \right)^\sigma ((1-\theta)Y_{t+1})^{-\sigma} + \beta(1-\delta) ((1-\theta)Y_{t+2})^{-\sigma} \alpha_2 A_{t+2}^{1-\sigma} \left(\frac{Y_{t+2}}{(1-\delta)\theta Y_t} \right)^\sigma \right]$$

Cancel all Y terms and the $(1-\theta)^{-\sigma}$ terms:

$$1 = \beta E \left[\alpha_1 A_{t+1}^{1-\sigma} \left(\frac{1}{\theta} \right)^\sigma + (1-\delta) \beta \alpha_2 A_{t+2}^{1-\sigma} \left(\frac{1}{(1-\delta)\theta} \right)^\sigma \right]$$

Shift θ to the LHS and move the expectations operator next to the A terms:

$$\theta^\sigma = \beta \alpha_1 E [A_{t+1}^{1-\sigma}] + (1-\delta)^{1-\sigma} \beta^2 \alpha_2 E [A_{t+2}^{1-\sigma}]$$

Using the definition of μ and isolating θ :

$$\theta = (\beta \alpha_1 \mu + (1-\delta)^{1-\sigma} \beta^2 \alpha_2 \mu)^{\frac{1}{\sigma}}$$

Full list of policy functions:

$$C_t = (1 - (\beta \alpha_1 \mu + (1-\delta)^{1-\sigma} \beta^2 \alpha_2 \mu)^{\frac{1}{\sigma}}) Y_t$$

$$K_{1,t+1} = I_t = ((\beta \alpha_1 \mu + (1-\delta)^{1-\sigma} \beta^2 \alpha_2 \mu)^{\frac{1}{\sigma}}) Y_t$$

$$K_{2,t+1} = (1-\delta) K_{1,t}$$

with

$$Y_t = A_t (\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 ((1-\delta) K_{1,t-1})^{1-\sigma} + (1 - \alpha_1 - \alpha_2))^{\frac{1}{1-\sigma}}$$

d) Discuss how, and why, a positive TFP shock affects consumption, investment and output in this model. How does the value of δ affect the predictions of the model?

Answer

Higher TFP raises the economy's production possibilities. Output rises directly, which also increases the marginal product of capital. The planner prefers a smooth path for consumption, so while some of the additional output is consumed, most of it is invested initially. The higher capital stock is then used to maintain higher consumption once the shock has dissipated.

Given the constant savings rate, a 1% shock to TFP will raise output, consumption and investment by 1%. The shock will generate persistent movements in the capital stock, so the effects on the economy will last longer than the shock itself.

A lower δ means that more of today's new capital remains useful tomorrow (as K_2). This increases the persistence of the effects of the shock. Higher δ therefore allows the model to generate additional endogenous internal persistence for output, investment and consumption.

e) Suppose, you solve a version of this model computationally using a linearization-based technique. Explain one advantage and one disadvantage of using this approach.

Answer

- Advantages: Fast and tractable; can handle a large number of variables and a large state space; allows for analytical insights; convenient calculation of impulse responses.
- Disadvantages: Accuracy is limited to local behavior around the steady state; the approximation may be poor for very non-linear models; hard to handle non-convexities.

Question 4 (20 points)

The U.S. government is currently considering large reductions in government spending, which could potentially lead to a lower public capital stock (for example, through reduced infrastructure spending). Using the New Keynesian model, this question asks you to analyze the macroeconomic impact, and how monetary policy might respond.

As usual, there are a continuum of identical households. The representative household makes consumption and labor supply decisions to maximize lifetime expected utility. In linearized form, the household first order conditions are as follows:

$$E_t \hat{c}_{t+1} - \hat{c}_t = \hat{i}_t - E_t \hat{\pi}_{t+1} \quad (4)$$

$$\hat{w}_t = \hat{c}_t + \psi \hat{n}_t \quad (5)$$

In percentage deviations from steady state: $\hat{c}_t$ is consumption, $\hat{w}_t$ is the real wage, and $\hat{n}_t$ is hours worked. In deviations from steady state: $\hat{i}_t$ is the nominal interest rate and $\hat{\pi}_t$ is inflation ($\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$; $\hat{p}_t$ is the percentage deviation of the price level from steady state). The coefficient $\psi = \frac{N}{1-N}$. N denotes steady state hours worked, $0 < N < 1$ and $\psi > 0$.²

The production side of the model is similar to the standard New Keynesian environment, except for the inclusion of public capital in the intermediate goods production process. Monopolistically competitive intermediate goods firms produce an intermediate good using labor as usual and public capital which is as an “external input” and not chosen by firms. The production function for product variety j is given by:

$$y_t(j) = K_{G,t}^\chi n_t(j) \quad (6)$$

Public capital $K_{G,t}$ is common across all firms and evolves as follows:

$$K_{G,t} = (1 - \delta)K_{G,t-1} + G_t \quad (7)$$

G_t is government investment. For simplicity, assume full depreciation $\delta = 1$ and $K_{G,t} = G_t$. χ is the elasticity of output with respect to public capital and $0 \leq \chi \leq 1$.

The rest of the model is standard and intermediate goods firms face a probability that they cannot adjust their price each period (the Calvo pricing mechanism). As usual, intermediate goods are aggregated into a final (homogenous) consumption good by final goods firms using the constant elasticity of substitution (CES) aggregator.

²Household preferences are $\eta \ln C_t + (1 - \eta) \ln(1 - N_t)$ where $0 < \eta < 1$.

The production side of the economy, when aggregated and linearized, can be described by the following set of linearized equilibrium conditions (the production function, the optimal hiring condition for labor and the dynamic evolution of prices):

$$\hat{y}_t = \chi \hat{k}_{G,t} + \hat{n}_t \quad (8)$$

$$\hat{w}_t = \hat{m}c_t + \chi \hat{k}_{G,t} \quad (9)$$

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda \hat{m}c_t \quad (10)$$

The resource constraint is:
$$\hat{y}_t = \gamma_c \hat{c}_t + \hat{g}_t \quad (11)$$

Monetary policy follows a simple Taylor Rule:
$$\hat{i}_t = \phi_\pi \hat{\pi}_t \quad (12)$$

Government investment (linearized) follows:
$$\hat{g}_t = \rho \hat{g}_{t-1} + e_t \quad (13)$$

e_t is i.i.d. $\hat{g}_t$ is the change in government investment relative to steady state as a proportion of steady state output: $\hat{g}_t = \frac{G_t - G}{Y}$. This implies that, with 100% depreciation each period, $\hat{k}_{G,t} = \frac{1}{\gamma_g} \hat{g}_t$ where γ_g is the steady state share of government investment in output and assume $0 < \gamma_g < 1$. Government investment is financed entirely by lump sum taxes on households. In percentage deviations from steady state: $\hat{m}c_t$ is real marginal cost, $\hat{y}_t$ is output and $\hat{k}_{G,t}$ is public capital. γ_c is the steady state share of consumption in output. λ is a function of model parameters, including the degree of price stickiness.³ Assume that $\phi_\pi > 1$, $0 < \rho < 1$.

a) Using the relevant equations above show that, with flexible prices, the natural level of output can be written as

$$\hat{y}_t^n = \Gamma \hat{g}_t \quad (14)$$

where

$$\Gamma \equiv \frac{1}{1 + \psi \gamma_c} \left(1 + \frac{\gamma_c(1 + \psi)\chi}{\gamma_g} \right)$$

Why does the response of the natural level of output depend on the value of χ ?

Answer

Start from:

$$\hat{w}_t = \hat{c}_t + \psi \hat{n}_t \quad (15)$$

and

$$\hat{w}_t = \hat{m}c_t + \chi \hat{k}_{G,t} \quad (16)$$

³ $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}$ where θ is the probability that a firm cannot adjust its price.

Furthermore, the natural level of output occurs under flexible prices, so $\hat{m}c_t = 0$. Because all firms are free to set the same price, there is no markup dispersion and $\hat{m}c_t = 0$. Also making use of the production function and the resource constraint yields:

$$\chi \hat{k}_{G,t} = \frac{1}{\gamma_c}(\hat{y}_t - \hat{g}_t) + \psi(\hat{y}_t - \chi \hat{k}_{G,t}) \quad (17)$$

Solving for $\hat{y}_t$ and putting a superscript n to denote the level of output under the assumption of flexible prices yields the result in the question:

$$\hat{y}_t^n = \Gamma \hat{g}_t \quad (18)$$

where

$$\Gamma \equiv \frac{1}{1 + \psi \gamma_c} \left(1 + \frac{\gamma_c(1 + \psi)\chi}{\gamma_g} \right)$$

Interpretation: The response of natural output depends on χ because public capital directly affects productivity. A higher χ means public capital is more productive, so changes in G_t have a larger effect on output. This makes public capital more like a TFP shock.

b) The full sticky price model can be simplified to 3 equations:

$$E_t \tilde{y}_{t+1} - \tilde{y}_t = \gamma_c(\hat{i}_t - E_t \hat{\pi}_{t+1} - \frac{1}{\gamma_c}(1 - \Gamma)(1 - \rho)\hat{g}_t) \quad (19)$$

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \kappa \tilde{y}_t \quad (20)$$

$$\hat{i}_t = \phi_\pi \hat{\pi}_t \quad (21)$$

plus the stochastic process for government investment (equation (13)). $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$ is the output gap and $\kappa = \frac{1 + \psi \gamma_c}{\gamma_c} \lambda$.

Using the method of undetermined coefficients, find the response of the output gap and inflation to an exogenous decrease in government investment when prices are sticky and monetary policy follows the Taylor Rule above. To do this, guess that the solution for each variable is a linear function of the government investment shock $\hat{g}_t$.

Answer

Using the guesses into the Phillips Curve:

$$\Lambda_\pi \hat{g}_t = (\beta \Lambda_\pi \rho + \kappa \Lambda_y) \hat{g}_t$$

The solution must satisfy:

$$\Lambda_\pi = \frac{\kappa}{1 - \beta \rho} \Lambda_y$$

Next, using the guesses, the AR(1) process and the policy rule in the dynamics IS curve:

$$\Lambda_y \hat{g}_t = \Lambda_y \rho \hat{g}_t - \gamma_c (\phi_\pi \Lambda_\pi \hat{g}_t - \Lambda_\pi \rho \hat{g}_t) - (\Gamma - 1)(1 - \rho) \hat{g}_t$$

Simplifying and substituting the result we found for Λ_π , the solution must satisfy:

$$\Lambda_y = \frac{(1 - \rho)(1 - \Gamma)}{1 - \rho + \Psi}$$

where

$$\Psi = \frac{\kappa \gamma_c (\phi - \rho)}{(1 - \beta \rho)}$$

and, from above,

$$\Lambda_\pi = \frac{\kappa}{1 - \beta \rho} \Lambda_y$$

At this stage, the effect on the output gap and inflation of a decrease in public investment is ambiguous. Note that the key question is whether Γ is less than one or not. If $\Gamma < 1$ these two expressions are positive. But Γ may also be greater than 1 depending on the value of χ .

c) Discuss how, and why, a fall in public investment affects the output gap and inflation in this model. Take care to explain how your answer depends on the value of χ (**Hint:** you may want to consider two cases: $\chi = 0$ vs. $\chi = 1$, and also consider the expression for Γ).

Answer

As noted in part (c), without some further assumptions on χ it is not possible to know whether the decrease in $\hat{g}_t$ raises or lowers the output gap and inflation. This shock has two different types of effects. First, this shock influences the economy in the same way as a pure government consumption shock. This can be seen when $\chi = 0$. In this case, the government spending shock is a pure demand shock (assuming the government spending is not useful, as is the case when $\chi = 0$). When prices are flexible all firms adjust prices and the natural level of output falls because households want to supply less labor. But when prices are sticky, firms who don't adjust prices lower production following the fall in demand. Labor demand therefore falls. Inflation also falls with the demand shock (some firms adjust prices), although sticker prices will tend to limit the effect on inflation. In addition to the neoclassical effects on the natural level, there is also a Keynesian-type demand effect. Because the output gap is negative, output falls by more in the sticky price model. Mathematically we can see this as follows. From the answers to parts (b) and (c), when $\chi = 0$, $\Gamma < 1$. The output gap and inflation always fall (under this monetary policy rule).

Now consider $\chi > 0$. This means the reduction in government spending influences productivity. In fact, because $K_{G,t} = G_t$, G_t (partly) shows up like a TFP shock.

We know that negative technology shocks lower the natural level of output but also generate rising prices. With price stickiness, output does not fall as much as it would if prices could rise freely. As a result, inflation and the output gap are positive. As χ gets larger, this effect starts to dominate the demand effect discussed above. For a large enough χ , $\Gamma > 1$ and inflation and the output gap become positive overall. Of course, the natural level of output still unambiguously decreases (as can be seen from the equation in part (b)). The effect on the natural level is increasing in χ because a higher χ implies a larger effect of public capital on firm productivity.

d) According to the monetary policy rule in equation (12), how will the monetary policymaker adjust the nominal interest rate following a fall in public investment? Does this result depend on the value of χ ? Provide some intuition for your result.

Answer

According to the policy rule in equation (12), the path of interest rates mirrors the path for inflation found in part (c). When χ is sufficiently large, the inflation response is positive and interest rates rise. This mirrors the result for a TFP shock in the standard model: lower TFP leads to rising interest rates, prices and the output gap. When χ is small, the demand effects start to dominate and prices and the output gap fall. In this case, interest rates also fall.

e) Now let's consider whether monetary policy could do better than in part (d). Can you think of a policy rule that would implement full stabilization of inflation and the output gap in this model? Explain. (**Hint:** you do not need to derive anything. You should be able to answer this from your knowledge of the model.)

Answer

From the 3-equation setup in part (b) you can see that this is still like a shock to the natural real rate of interest, although it has elements more like a demand shock and other elements more like a TFP shock. It is potentially feasible to fully stabilize the output gap and inflation by setting the real interest rate equal to the natural real interest rate. Simply following this rule would not deliver a unique equilibrium, so the rule needs to be augmented by an inflation feedback term. A rule such as:

$$\hat{i}_t = \hat{r}_t^n + \phi \hat{\pi}_t$$

with $\phi > 1$ would implement full stabilization of inflation and the output gap as the unique equilibrium.